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Institute Name:	Silver Oak College of Computer Application		
Programme Name:	Master of Science Data Science	Semester:	1
Course Code:	DSM5061C	Course Name:	Information and Data Security
Unit No.:	2	Unit Name:	Data Privacy and Protection

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Introduction to Data Privacy and Protection

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Data privacy governs how personal information is collected and used; data protection safeguards it against loss, misuse, or unauthorized access.

Privacy

The right to control one's own personal data

Protection

Technical & organizational safeguards

Compliance

Alignment with IT Act, GDPR & DPDP Act

Personal Data, Sensitive Personal Data and Personally Identifiable Information (PII)

Personal Data

Any data relating to an identifiable person

Sensitive Personal Data

Health, biometric, financial, religious data

PII

Data that can directly identify an individual

Examples: Name & address (Personal Data) • Medical records & biometrics (Sensitive) • Aadhaar, PAN, email (PII)

Data Privacy Principles and Privacy Requirements

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Lawfulness & Fairness

Process data with valid legal basis

Purpose Limitation

Use data only for stated purposes

Data Minimization

Collect only what is necessary

Accuracy

Keep data correct and up to date

Storage Limitation

Retain data only as long as needed

Accountability

Demonstrate compliance & governance

Data Collection and Secure Data Acquisition

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Consent

Obtain explicit user consent

Collect

Gather data via secure channels

Validate

Verify accuracy & completeness

Secure Transfer

Encrypt data in transit (TLS)

Secure acquisition ensures data is captured lawfully, minimally, and transmitted through encrypted, tamper-resistant channels.

Data Storage and Secure Data Management

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Encryption at Rest

Data encrypted on disk & databases

Access Control

Role-based restricted access

Redundancy

Replicated storage for resilience

Monitoring

Continuous audit of access logs

Data Processing and Data Sharing

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Internal Processing → Authorized Sharing → 3rd-Party Agreement

- Processing must align with the original purpose of collection
- Sharing requires data processing agreements (DPAs) with third parties
- Cross-border transfers need adequacy or contractual safeguards

Data Disposal and Secure Data Destruction

Retention Ends

Data reaches end of lifecycle

Secure Wipe

Overwrite / cryptographic erasure

Physical Destruction

Shred or degauss media

Certify

Issue certificate of destruction

Improper disposal is a leading cause of data breaches — destruction must be verifiable, irreversible, and documented.

Data Classification and Data Categorization

Classification determines the handling, access, and protection controls applied to each data category.

Restricted	Highest sensitivity — strict controls
Confidential	Restricted to authorized roles
Internal	Limited to organization
Public	No harm if disclosed

Data Masking and Data Protection Techniques

`Original: John Smith, 4111-2233-4455-6678`

`J*** S****, XXXX-XXXX-XXXX-6678`

Encryption

Reversible with a key

Tokenization

Replace with non-sensitive token

Redaction

Permanently remove/black-out

Data Anonymization and Pseudonymization

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Anonymization

- Data permanently stripped of identifiers
- Cannot be re-linked to an individual
- Falls outside most privacy regulations

Pseudonymization

- Identifiers replaced with reversible tokens
- Re-identification possible with a separate key
- Still considered personal data under law

Secure File Sharing and Secure Data Storage

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Encrypted Links

Password-protected share links

Access Expiry

Time-limited download windows

Secure Cloud Storage

Encrypted, access-audited repositories

Least Privilege

Share only with authorized recipients

The 3-2-1 Backup Rule

3
Copies of Data
One primary + two backups

2
Storage Media Types
e.g. disk and cloud

1
Off-site Copy
Protects against site disasters

Business Continuity Planning (BCP) ensures rapid recovery of critical data and services after disruption, minimizing downtime and data loss.

Digital Footprint and Social Media Privacy

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Active Footprint

- Posts, comments, uploaded content
- Deliberately shared information
- Directly controllable by the user

Passive Footprint

- Cookies, location, browsing history
- Collected without explicit action
- Requires privacy settings & controls

Data Privacy Challenges in Digital Analytics

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Tracking Cookies

Persistent cross-site user tracking

Behavioral Profiling

Inferred preferences from activity

Third-Party Data

Data shared with ad/analytics networks

Consent Gaps

Analytics running before consent capture

Data Protection Policies, Handling Best Practices and Data Loss Prevention (DLP) Awareness

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Written Policies

Documented data handling standards

Best Practices

Least privilege, encryption, training

DLP Monitoring

Detect & block unauthorized data flow

Incident Response

Defined breach reporting procedure

DLP tools continuously scan endpoints, networks, and storage to prevent sensitive data from leaving the organization without authorization.

Thank you